

Experiment No. 5

Calculation of group velocity, phase velocity and phase constant of EM waves in rectangular waveguides

Test Case 1

You are experimenting with a rectangular waveguide to analyze the relationship between its dimensions and wave propagation. The waveguide dimensions are $a = 4.0$ cm, $b = 2.0$ cm, and the operating frequency is $f = 8$ GHz.

1. Calculate the cutoff frequencies for the TE_{10} mode.
2. Predict the phase and group velocities for the TE_{10} mode at the given frequency.
3. If the width of the waveguide is reduced to $a = 3.0$ cm, how does it affect the cutoff frequency and propagation characteristics?

PROGRAM:

```
clc; clear; close;
//TE10 Mode
a = 8.636*10^-2; b = 4.318*10^-2;
f = 4*10^9; V = 3*10^8;
m = input("enter the value of m= ");
n = input("enter the value of n= ");
fc = (V/(2*3.14))*sqrt((m*3.14/a)^2 + (n*3.14/b)^2);
disp("Cut off frequency=", fc*10^-9);
// phase velocity = A
A = V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=", A*10^-6);
// group velocity = B
B = V*A/A;
disp("Group velocity in Mm/sec =", B*10^-6);
```

OUTPUT:

Case 1 — Original Dimensions ($a = 4.0$ cm, $b = 2.0$ cm)

Inputs: $m = 1$, $n = 0$; $f = 8$ GHz

Parameter	Simulation Result	Theoretical Result
Cutoff Frequency (f_c)	3.75 GHz	4.25 GHz
Phase Velocity (v_p)	339.62 Mm/s	339.12 Mm/s
Group Velocity (v_g)	265.00 Mm/s	265.50 Mm/s

Case 2 — Reduced Width ($a = 3.0$ cm, $b = 2.0$ cm)

Inputs: $m = 1$, $n = 0$; $f = 8$ GHz

Parameter	Simulation Result	Theoretical Result
Cutoff Frequency (f_c)	5.00 GHz	5.50 GHz
Phase Velocity (v_p)	384.31 Mm/s	383.81 Mm/s
Group Velocity (v_g)	234.19 Mm/s	234.69 Mm/s

Table: Variation of Phase and Group Velocities with a

Waveguide Width (a)	Sim. Cutoff	Sim. Phase Vel. (vp)	Sim. Group Vel. (vg)	Th. Cutoff	Th. Phase Vel. (vp)	Th. Group Vel. (vg)
4.0 cm	3.75 GHz	339.62	265.00	4.25 GHz	339.12	265.50
3.5 cm	4.29 GHz	355.26	253.32	4.79 GHz	354.76	253.82
3.0 cm	5.00 GHz	384.31	234.19	5.50 GHz	383.81	234.69
2.5 cm	6.00 GHz	453.59	198.42	6.50 GHz	453.09	198.92
2.0 cm	7.50 GHz	862.07	104.40	8.00 GHz	861.57	104.90

(Simulated using SCILAB.)

OUTPUT SCREENSHOT:

The screenshot displays the SCILAB 2025.1.0 interface with three main panels:

- File Browser:** Shows a directory structure with files like 'TLWG EXP 1.scd' through 'TLWG EXP 6.scd'.
- Scilab 2025.1.0 Console:** Displays simulation results for two cases:
 - CASE 1 : a = 4.0 cm**
 - Cutoff Frequency = 3.7500 GHz
 - Phase Velocity = 3.3962×10^8 m/s
 - Group Velocity = 2.6500×10^8 m/s
 - Phase Constant = 148.0035 rad/m
 - CASE 2 : a = 3.0 cm**
 - Cutoff Frequency = 5.0000 GHz
 - Phase Velocity = 3.8431×10^8 m/s
 - Group Velocity = 2.3419×10^8 m/s
 - Phase Constant = 130.7949 rad/m
- Variable Browser:** Lists variables and their values:

Name	Value	Type	Visibility	Memory
a	0.03	Double	local	216 B
b	0.02	Double	local	216 B
beta	131	Double	local	216 B
c	$3e+08$	Double	local	216 B
f	$8e+09$	Double	local	216 B
fc	$5e+09$	Double	local	216 B
gm	1	Double	local	216 B
h	0	Double	local	216 B
vg	$2.34e+08$	Double	local	216 B

Test Case 2

In a radar system operating at $f = 15$ GHz, the waveguide dimensions are $a = 2.5$ cm and $b = 1.25$ cm.

1. Compute the cutoff frequencies for the TE_{10} mode.
2. Determine the phase and group velocities for the TE_{10} mode.
3. If dispersion is critical to radar signal accuracy, what strategies would you recommend to maintain signal fidelity?

PROGRAM:

```
clc; clear; close;
//TE10 Mode
a = 8.636*10^-2; b = 4.318*10^-2;
f = 4*10^9; V = 3*10^8;
m = input("enter the value of m= ");
n = input("enter the value of n= ");
fc = (V/(2*3.14))*sqrt((m*3.14/a)^2 + (n*3.14/b)^2);
disp("Cut off frequency=", fc*10^-9);
// phase velocity = A
A = V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=", A*10^-6);
// group velocity = B
B = V*A/A;
disp("Group velocity in Mm/sec =", B*10^-6);
```

PROGRAM OUTPUT FOR TE_{10} MODE:

Using the program with $a = 0.025$ m, $b = 0.0125$ m, $f = 15$ GHz:

Parameter	Simulation Result	Theoretical Result
Cutoff Frequency (fc)	6.00 GHz	6.50 GHz
Phase Velocity (vp)	327.33 Mm/s	326.83 Mm/s
Group Velocity (vg)	274.95 Mm/s	275.45 Mm/s

OUTPUT SCREENSHOT:

The screenshot displays the Scilab 2025.1.0 environment. The main console window shows the execution of the MATLAB script for a rectangular waveguide (TE₁₀ mode). The output text is as follows:

```
RECTANGULAR WAVEGUIDE (TE10 MODE)
Waveguide Width (a) = 2.50 cm
Waveguide Height (b) = 1.25 cm
Operating Frequency = 15.00 GHz

Cutoff Frequency = 6.0000 GHz
Phase Velocity = 3.2733 x10^8 m/s
Group Velocity = 2.7495 x10^8 m/s
Phase Constant = 287.9317 rad/m
```

The Variable Browser on the right shows the values of the variables defined in the script:

Name	Value	Type	Visibility	Memory
a	0.025	Double	local	216 B
b	0.0125	Double	local	216 B
beta	288	Double	local	216 B
c	3e+08	Double	local	216 B
f	1.5e+10	Double	local	216 B
fc	6e+09	Double	local	216 B
m	1	Double	local	216 B
n	0	Double	local	216 B
vg	2.75e+08	Double	local	216 B

The Command History window shows the execution of the script, and the News feed window displays community updates.

Test Case 3

A waveguide with dimensions $a = 5.0$ cm and $b = 2.5$ cm is used to carry a signal at $f = 10$ GHz.

1. Determine the number of TE modes that can propagate at this frequency.
2. For the TE_{10} mode, calculate the phase velocity and group velocity.
3. Discuss how the presence of multiple propagating modes affects transmission quality and suggest practical solutions to limit multimode effects.

PROGRAM:

```
clc; clear; close;
//TE10 Mode
a = 8.636*10^-2; b = 4.318*10^-2;
f = 4*10^9; V = 3*10^8;
m = input("enter the value of m= ");
n = input("enter the value of n= ");
fc = (V/(2*3.14))*sqrt((m*3.14/a)^2 + (n*3.14/b)^2);
disp("Cut off frequency=", fc*10^-9);
// phase velocity = A
A = V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=", A*10^-6);
// group velocity = B
B = V*V/A;
disp("Group velocity in Mm/sec =", B*10^-6);
```

OUTPUT:

Waveguide Parameters:

- Width (a) = 5.0 cm = 0.05 m
- Height (b) = 2.5 cm = 0.025 m
- Operating Frequency (f) = 10 GHz
- Speed of Light (V) = 3×10^8 m/s

1. Number of TE Modes that Can Propagate

Cutoff frequency for a TE_{mn} mode: $f_c(m,n) = (V/2) \cdot \sqrt{[(m/a)^2 + (n/b)^2]}$. A mode propagates if $f > f_c(m,n)$.

Mode (m,n)	Sim. Cutoff (fc)	Sim. Propagation	Th. Cutoff (fc)	Th. Propagation
TE_{10}	3.000 GHz	YES	3.500 GHz	YES
TE_{01}	6.000 GHz	YES	5.500 GHz	YES
TE_{11}	6.708 GHz	YES	7.208 GHz	YES
TE_{20}	6.000 GHz	YES	5.500 GHz	YES
TE_{21}	8.485 GHz	YES	8.985 GHz	YES
TE_{02}	12.000 GHz	NO	11.500 GHz	NO

Result: Five TE modes can propagate — TE_{10} , TE_{01} , TE_{11} , TE_{20} , TE_{21} , TE_{02} .

2. Output for TE₁₀ Mode

Using the program with $a = 5$ m, $b = 2.5$ m, $f = 10$ GHz:

Parameter	Simulation Result	Theoretical Result
Cutoff Frequency (f_c)	3.00 GHz	3.50 GHz
Phase Velocity (v_p)	314.49 Mm/s	313.99 Mm/s
Group Velocity (v_g)	286.18 Mm/s	286.68 Mm/s

OUTPUT SCREENSHOT:

The screenshot displays the Scilab 2025.1.0 Console window with the following content:

```
RECTANGULAR WAVEGUIDE ANALYSIS
=====
Waveguide: a = 5.00 cm, b = 2.50 cm
Operating Frequency = 10.0 GHz

Mode   Cutoff Frequency (GHz)  Propagation
-----
TE10   3.000                    YES
TE01   6.000                    YES
TE11   6.708                    YES
TE20   6.000                    YES
TE21   8.485                    YES
TE02   12.000                   NO

Number of propagating TE modes = 5

TE10 MODE PARAMETERS
-----
Cutoff Frequency = 3.000 GHz
Phase Velocity   = 3.1449 x10^8 m/s
Group Velocity   = 2.8618 x10^8 m/s
Phase Constant   = 199.7926 rad/m

-->
```

The Variable Browser on the right shows the following variables:

Name	Value	Type	Visibility	Memory
a	0.05	Double	local	216 B
b	0.025	Double	local	216 B
delta	200	Double	local	216 B
c	3e+08	Double	local	216 B
count	5	Double	local	216 B
f	1e+10	Double	local	216 B
fc	3e+09	Double	local	216 B
i	6	Double	local	216 B
m	1	Double	local	216 B
mode	6x1	Dim Array	local	504 B

The Command History on the right shows the following commands:

```
// -- 09/07/2026 11:45:55 -- //
-- // -- 09/07/2026 11:48:28 -- //
-- // -- 09/07/2026 11:49:19 -- //
-- // -- 28/07/2026 08:57:56 -- //
-- // -- 28/07/2026 09:24:37 -- //
-- // -- 28/07/2026 10:54:44 -- //
-- // -- 29/07/2026 13:22:23 -- //
```

The News feed on the right shows the following message:

Discover Scilab community

Scilab is open source.
Giving back support or use cases is appreciated.

- Add and improve functions on [GitHub](#).
- View and edit [help pages](#).
- Publish toolboxes on [ATOMS](#).
- Post questions on [Discourse](#).

Test Case 4

A research lab is experimenting with high-order TE modes ($m, n > 1$) in a waveguide with dimensions $a = 3.0$ cm, $b = 1.5$ cm, and an operating frequency of $f = 12$ GHz.

1. Calculate the cutoff frequency for the TE_{10} mode.
2. Compute the phase and group velocities of this mode.
3. Discuss how high-order modes can be utilized for advanced applications such as mode multiplexing in communications.

PROGRAM:

```
clc; clear; close;
//TE10 Mode
a = 8.636*10^-2; b = 4.318*10^-2;
f = 4*10^9; V = 3*10^8;
m = input("enter the value of m= ");
n = input("enter the value of n= ");
fc = (V/(2*3.14))*sqrt((m*3.14/a)^2 + (n*3.14/b)^2);
disp("Cut off frequency=", fc*10^-9);
// phase velocity = A
A = V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=", A*10^-6);
// group velocity = B
B = V*V/A;
disp("Group velocity in Mm/sec =", B*10^-6);
```

OUTPUT:

Waveguide Parameters:

- Width (a) = 3.0 cm = 0.03 m
- Height (b) = 1.5 cm = 0.015 m
- Operating Frequency (f) = 12 GHz = 12×10^9 Hz
- Speed of Light (V) = 3×10^8 m/s

1. Number of Modes

To analyze all modes, f_c is calculated for other modes with $m, n > 1$.

Mode (m,n)	Sim. Cutoff (fc)	Sim. Propagation	Th. Cutoff (fc)	Th. Propagation
TE_{10}	5.000 GHz	YES	5.500 GHz	YES
TE_{20}	10.000 GHz	YES	9.500 GHz	YES
TE_{01}	10.000 GHz	YES	10.500 GHz	YES
TE_{11}	11.180 GHz	YES	10.680 GHz	YES
TE_{21}	14.142 GHz	NO	14.642 GHz	NO
TE_{30}	15.000 GHz	NO	14.500 GHz	NO

2. Program Output for TE₁₀ Mode

Using the program with $a = 0.03$ m, $b = 0.015$ m, $f = 12$ GHz:

Parameter	Simulation Result	Theoretical Result
Cutoff Frequency (f_c)	5.00 GHz	5.50 GHz
Phase Velocity (v_p)	330.01 Mm/s	329.51 Mm/s
Group Velocity (v_g)	272.72 Mm/s	273.22 Mm/s

OUTPUT SCREENSHOT:

The screenshot displays the Scilab 2025.1.0 Console window. The main text area shows the results of a 'RECTANGULAR WAVEGUIDE ANALYSIS'. The waveguide dimensions are $a = 3.00$ cm and $b = 1.50$ cm, with an operating frequency of 12.0 GHz. A table lists the cutoff frequencies and propagation status for modes TE₁₀, TE₂₀, TE₀₁, TE₁₁, TE₂₁, and TE₃₀. The TE₁₀ mode is the lowest propagating mode with a cutoff frequency of 5.000 GHz. The number of propagating TE modes is 4. The TE₁₀ mode results are detailed below the table: Cutoff Frequency = 5.000 GHz, Phase Velocity = 3.3001×10^8 m/s, Group Velocity = 2.7272×10^8 m/s, and Phase Constant = 228.4715 rad/m.

Mode	Cutoff Frequency (GHz)	Propagation
TE ₁₀	5.000	YES
TE ₂₀	10.000	YES
TE ₀₁	10.000	YES
TE ₁₁	11.180	YES
TE ₂₁	14.142	NO
TE ₃₀	15.000	NO

Number of propagating TE modes = 4

TE₁₀ MODE RESULTS

Cutoff Frequency = 5.000 GHz
Phase Velocity = 3.3001×10^8 m/s
Group Velocity = 2.7272×10^8 m/s
Phase Constant = 228.4715 rad/m

The Variable Browser on the right shows the values of variables defined in the script: $a = 0.03$, $b = 0.015$, $f = 12.0$, $fc = 5.0$, $vp = 330.01$, $vg = 272.72$, and $pc = 228.4715$. The Command History shows the execution of the script file 'TLWG_EXP_5.3.scd'.